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| **Time taken** | 10 secs |
| **Marks** | 0.00/53.00 |
| **Grade** | **0.00** out of 100.00 |

Question **1**

Not answered

Marked out of 1.00

Flag question

John has depressive disorder. He is skeptical about taking antidepressants as he had sexual side effects with them. Which of the following has the lowest risk of sexual dysfunction?

Select one:

- Mirtazapine
- Venlafaxine
- Agomelatine
- Trazadone
- Sertraline

Your answer is incorrect.

With Agomelatine, there is no clear effect on orgasm, erection or libido. The incidence of sexual dysfunction is similar to placebo.

The correct answer is: Agomelatine

Question **2**

Not answered

Marked out of 1.00

Flag question

Paul has been diagnosed with frontotemporal dementia. He has a strong family history. A mutation on which of the following chromosome predisposes a person to develop this condition?

Select one:

- 7
- 19
- 5
- 17
- 22

Your answer is incorrect.

Frontotemporal lobar degeneration (FTLD) refers to the 3 different syndrome of frontotemporal dementia (FTD), progressive non fluent aphasia and semantic dementia. ?Some patients with FTLD show tau protein based pathological changes. In familial cases, mutations have been identified in the

microtubule-associated protein tau gene (MAPT) on chromosome 17q21.

The correct answer is: 17

Question **3**

Not answered

Marked out of 1.00

Flag question

Which of the following deletion syndromes is considered as the highest known genetic risk factor for schizophrenia other than having a monozygotic twin sibling with the illness?

Select one:

- 22 q 11.2
- 14 q 11.2
- 19 q 11.2
- 25 q 11.2
- 5q 11.2

Your answer is incorrect.

People with 22q11.2 deletion syndrome (velo-cardio-facial syndrome) have a 30-fold risk of developing schizophrenia. During adolescence and early adulthood, up to 30% of people with the syndrome develop schizophrenia, thus suggesting that the 22q11.2 deletion can be considered the highest known genetic risk factor for schizophrenia other than having a monozygotic twin sibling with the illness.

The correct answer is: 22 q 11.2

Question **4**

Not answered

Marked out of 1.00

Flag question

Huntington's disease is a progressive neurodegenerative disorder, which is characterized by;

Select one:

- CTG > 35 expansion
- CCA > 35 expansion
- CCA < 35 expansion
- CAG< 35 expansion
- CAG > 35 expansion

Your answer is incorrect.

Huntington's disease (HD) is a progressive neurodegenerative disorder with autosomal-dominant inheritance. The disease is caused by a CAG trinucleotide repeat expansion located in the first exon of the HD gene. The CAG repeat is highly polymorphic and varies from 6 to 37 repeats on chromosomes of unaffected individuals and from more than 30 to 180 repeats on chromosomes of HD patients. (Ref: Möncke-Buchner E, Reich S, Mücke M, et al. Counting CAG repeats in the Huntington's disease gene by restriction endonuclease EcoP15I cleavage. Nucleic Acids Research. 2002;30(16):e83)

The correct answer is: CAG > 35 expansion

Question **5**

The gene most strongly implicated in ADHD is

Not answered

Marked out of 1.00

Flag question

Select one:

- Dopamine 4 receptor gene
- Dopamine 5 receptor gene
- Dopamine 1 receptor gene
- Dopamine 2 receptor gene
- Dopamine 3 receptor gene

Your answer is incorrect.

The gene most strongly implicated in ADHD is the D4 dopamine receptor gene DRD4. The distribution of DRD4 mRNA in the brain suggests that it plays a role in cognitive and executive functions. These functions are implicated in the pathophysiology of ADHD (Patterson 1999)

The correct answer is: Dopamine 4 receptor gene

Question **6**

Not answered

Marked out of 1.00

Flag question

Mutations in the progranulin gene are known to cause;

Select one:

- Lewy body dementia
- Parkinson's disease
- CADASIL
- Frontotemporal dementia
- CreutzfeldtJakob disease

Your answer is incorrect.

Mutations in the progranulin gene are known to cause frontotemporal lobar degeneration, leading to dementia (autosomal dominant type).

The correct answer is: Frontotemporal dementia

Question **7**

Not answered

Marked out of 1.00

Flag question

CADASIL is associated with mutation in which of the following genes?

Select one:

- COMT
- Dysbindin
- LRRK2
- Notch3
- Parkin gene

Your answer is incorrect.

An unusual form of subcortical small vessel disease occurs in the rare inherited condition of CADASIL (cerebral autosomal dominant arteriopathy with subcortical infarcts and leuko-encephalopathy) in which a mutation in the Notch3 gene results in deposition of granular material in the walls of small arteries and arterioles, narrowing of vessel lumens and destruction of smooth muscle cells in their walls. (Ref: Vascular dementia: a pragmatic review; BJPsych Advances; Sept 2012;372-380)

The correct answer is: Notch3

Question **8**

Not answered

Marked out of 1.00

Flag question

Which of the following structures is involved in the pathophysiology of Wernicke's encephalopathy?

Select one:

- Amygdala
- Mamillary bodies
- Hippocampus
- Prefrontal cortex
- Basal ganglia

Your answer is incorrect.

In Wernicke's encephalopathy, pathological lesions are seen in mammillary bodies, thalamic nuclei and the walls of the third ventricle.

The correct answer is: Mamillary bodies

Question **9**

Not answered

Marked out of 1.00

Flag question

Thomas Wilkins is a 56-year-old man who presented to his GP with some personality changes, including anxiety, depression, confusion and memory loss. His movements lacked coordination and he began to experience falls. There was a suspicion of vCJD and a head MRI was arranged. Which of the following signs gives an accurate diagnostic sign for vCJD (variant Creutzfeldt-Jakob disease)?

Select one:

- Frontal lobe atrophy
- Hypothalamus hyperintensity
- Pulvinar sign
- Pineal gland sign
- Cerebellar hypointensity

Your answer is incorrect.

The Pulvinar sign (symmetrical hyperintensity in the pulvinar (posterior) nuclei of the thalamus) on brain MR images is a highly accurate sign for vCJD.

The correct answer is: Pulvinar sign

Question **10**

The Nerve Supply of Superior Oblique muscle is;

Not answered

Marked out of 1.00

Flag question

Select one:

- Abducent
- Trochlear
- Trigeminal
- Oculomotor
- Facial

Your answer is incorrect.

The Superior oblique is the only extra ocular muscle innervated by the trochlear nerve.

The correct answer is: Trochlear

Question **11**

Not answered

Marked out of 1.00

Flag question

Mr Cox is a 56-year-old man with chronic schizophrenia, who has been smoking 20 cigarettes a day since his teenage years. On a routine medical examination, it is noted that the BP in his right arm is 184/98, and on the left 155/80. The examination is repeated on several other occasions, and the difference in blood pressure between the arms is found to be consistent. Which artery is most likely to be affected?

Select one:

- Radial artery
- Cerebellar artery
- Subclavian artery
- Posterior inferior cerebral artery
- Middle cerebral artery

Your answer is incorrect.

The vignette suggests that the subclavian artery is affected, most likely by atherosclerosis. Discrepancy in left and right arm blood pressures is associated with higher risk of cardiovascular events.

The correct answer is: Subclavian artery

Question **12**

Not answered

Marked out of 1.00

Flag question

Tom has severe anxiety disorder and recently started on Pregabalin. His partner, who is a chemistry graduate, is keen about its receptor action. Pregabalin acts on which of the following receptors?

Select one:

- GABA
- Glutamate
- Acetyl choline
- Noradrenaline
- Serotonin

Your answer is incorrect.

Pregabalin is a GABA analogue structurally (similar to gabapentin). Like gabapentin, pregabalin binds to the a2d subunit of the voltage-dependent calcium channel in the central nervous system. This may subtly reduce release ofcertain neurotransmitters. It may as well influence GABA-ergic neurotransmission. It has anti-epileptic, analgesic (neuropathic pain) and anxiolytic effects. It is more potent than gabapentin hence has a higher therapeutic index and fewer dose-related side effects.

The correct answer is: GABA

Question **13**

Not answered

Marked out of 1.00

Flag question

Benzodiazepines enhance the effects of which of the following neurotransmitters?

Select one:

- Acetyl choline
- GABA
- Serotonin
- Norepinephrine
- Dopamine

Your answer is incorrect.

It is widely believed that benzodiazepines enhance the effects of the neurotransmitter gamma-aminobutyric acid (GABA). They do so through action at benzodiazepine receptors.

The correct answer is: GABA

Question **14**

Not answered

Marked out of 1.00

Flag question

Casper Williams is a 25-year-old postgraduate student studying business administration. He attends the student mental health service complaining of memory loss, which is affecting his ability to study. He worries that he will fail his MBA. On exploration of his social history, he admits to frequently using angel dust. Phencyclidine-induced amnesia is due to action on which of the following receptors?

Select one:

- Dopamine
- Serotonin
- Glutamate
- Noradrenaline
- Acetyl choline

Your answer is incorrect.

Phecyclidine (PCP, angel dust) primarly acts as an antagonist at ionotropic glutamate receptors such as NMDA.

The correct answer is: Glutamate

Question **15**

Not answered

Marked out of 1.00

Flag question

Phencyclidine acts on which receptor to produce symptoms of schizophrenia?

Select one:

- Kainate
- NMDA receptor
- AMPA
- Dopamine receptor
- Serotonin receptor

Your answer is incorrect.

Phencyclidine blocks the activity of the NMDA receptor. It is a NMDA receptor antagonists that induces both positive and negative symptoms of Schizophrenia in healthy volunteers. This is possibly due to the modulation of the dopamine system and therefore exacerbates symptoms of patients with Schizophrenia.

The correct answer is: NMDA receptor

Question **16**

Not answered

Marked out of 1.00

Flag question

Which of the following statements about dopamine D2 receptors is NOT true?

Select one:

- They act via the tuberoinfundibular pathway
- They are excitatory
- They are G protein-coupled
- They constitute both pre and post-synaptic receptors
- They affect adenylyl cyclase activity

Your answer is incorrect.

D2 are inhibitory receptors that have both pre- and post-synaptic functions. They inhibit adenylyl cyclase activity and act via the tuberoinfundibular pathways that inhibit prolactin. Antipsychotics remove the inhibitory control of dopamine by blocking D2 receptors in tuberoinfundibular tract. This leads to hyperprolactinaemia, which presents asgynaecomastia in males and galactorrhoea in females. Long-standing prolactin increase may lead to osteoporosis.

The correct answer is: They are excitatory

Question **17**

Not answered

Marked out of 1.00

Flag question

Lucy is taking antipsychotic medications and complained of sexual dysfunction, menstrual disturbances and galactorrhoea. Which dopamine pathway when affected by antipsychotic drugs can induce such symptoms?

Select one:

- Niagrostriatal pathway
- Mesocortical pathway
- Tuberoinfundibular pathway

- All of the above
- Mesolimbic pathway

Your answer is incorrect.

Lucy is exhibiting features of raised prolactin levels. Hyperprolactinaemia is associated with a number of adverse consequences such as sexual dysfunction, menstrual disturbances, breast growth, galactorrhoea and reductions in bone mineral density. A reduction in the dopaminergic input to the lactotroph cells results in a rapid increase in prolactin secretion. Such a reduction in dopamine can occur through the administration of antipsychotics, which acts on dopamine receptors, especially D2 in the tuberoinfundibular pathway of the brain.

The correct answer is: Tuberoinfundibular pathway

Question **18**

Not answered

Marked out of 1.00

Flag question

Mrs.Mokolo has been reading up on the biochemistry of SSRIs, after you started her on Sertraline for an episode of depression. She reminds you that the serotonergic receptor with auto-regulatory function in synaptosomes (5-HT terminals) is:

Select one:

- 5-HT1B
- 5-HT2B
- 5-AT1A
- 5-HT1D
- 5-HT2A

Your answer is incorrect.

It is the 5-HT1B auto-receptors that are located in serotonin terminals. 5-HT1B auto-receptors regulate 5-HT reuptake, as well as modulating 5-HT release and synthesis.

The correct answer is: 5-HT1B

Question **19**

Not answered

Marked out of 1.00

Flag question

Toby is a 12-year-old boy in his first year of secondary school. His form tutor complains that he is extremely disruptive in school; jumping on desks and pushing in the cafeteria queue. You have reviewed him in your outpatient clinic and believe him to be suffering from ADHD. You arrange a meeting with his parents to discuss commencing medications. What is the mechanism of action of methylphenidate in treating ADHD?

Select one:

- Serotonin re-uptake inhibition
- Norepinephrine–Dopamine reuptake inhibitor
- Sigma-1 antagonist
- Dopamine agonist
- Catecholamine re-uptake inhibition

Your answer is incorrect.

Methylphenidate is a phenethylamine stimulant that binds and blocks norepinephrine and dopamine transporters. The levels of these two hormones are thought to be low in the pre-frontal cortex of individuals with ADHD and, as a consequence, affect executive function (inhibiting behaviours, reasoning, organizing etc). Methylphenidate increases the firing rate of norepinephrine and dopamine, which ultimately enhances executive functioning.

The correct answer is: Norepinephrine–Dopamine reuptake inhibitor

Question **20**

Not answered

Marked out of 1.00

Flag question

Which of the following antipsychotic is a highly selective dopamine antagonist?

Select one:

- Sulpride
- Olanzapine
- Clozapine
- Haloperidol
- Chlorpromazine

Your answer is incorrect.

Amisulpride and Sulpride have highly selective dopamine antagonistic property.

The correct answer is: Sulpride

Question **21**

Not answered

Marked out of 1.00

Flag question

James comes to your memory clinic with a two-month history of short-term memory loss. He is concerned that he has Parkinson's disease as his late father had suffered from the disease. Which of the following genes is associated with Parkinson's disease?

Select one:

- Alpha synuclein
- Neuregulin
- Presenilin-2
- Amyloid Precursor Protein
- Dysbindin

Your answer is incorrect.

Mutations in the following genes; alpha-synuclein (SNCA), parkin (PRKN), leucine-rich repeat kinase 2 (LRRK2), PTEN-induced putative kinase 1 (PINK1) are known to cause Parkinson's disease.

The correct answer is: Alpha synuclein

Question **22**

Not answered

You are asked by the matron in A&E to review Sarah Moore, a 43-year-old mother of two, who separated from her husband, John, earlier this year. For the past few months, Sarah has been spreading

Marked out of 1.00

Flag question

lies about John to other parents, including mistruths that he is taking their two children out of school during school hours. This resulted in a meeting between John and the school welfare officer. Sarah appears happy at this conclusion and tells you that it proves she is the better parent. These false accounts appear to coincide with John's new relationship. What is the name given to this particular phenomenon?

Select one:

- Falsification
- Pseudologia fantastica
- Confabulation
- Déjà vu
- Depersonalisation

Your answer is incorrect.

Pseudologiafantastica is the production of convincing false accounts often with apparent sincere conviction. There may be a grandiose or over-exaggerated flavour to the accounts produced. It is a feature of Munchausen's disease. (Ref. Oxford Handbook of Psychiatry). Their (incessant) lies are mostly aimed at attracting noise and inflating their importance (Delhi Psychiatry Journal 2007)

The correct answer is: Pseudologia fantastica

Question **23**

Not answered

Marked out of 1.00

Flag question

Brian Hughes is a war veteran who has been diagnosed with PTSD. He reports that his sleep is particularly disrupted and undergoes investigation at a sleep clinic. Which of the following findings is most consistent with sleep and PTSD?

Select one:

- Decreased stage 1 NREM sleep
- Increased frequency of rapid eye movements in REM sleep
- Longer total sleep time
- Decreased sleep onset latency
- Longer duration of stage 2 NREM sleep

Your answer is incorrect.

The effect on sleep in people with PTSD is the focus of much research and there have been several findings; the most consistent is that in those with PTSD (compared with controls), there is an increase in stage 1 NREM sleep, decreased Slow Wave Sleep (SWS) and increased average number of rapid eye movements per minute in REM sleep

The correct answer is: Increased frequency of rapid eye movements in REM sleep

Question **24**

Not answered

Marked out of 1.00

What type of EEG changes are routinely seen in Delirium Tremens?

Select one:

- Focal delta activity.

Flag question

- Increased triphasic waves
- Low voltage fast activity
- Diffuse slowing of the background rhythm
- Generalized periodic 1-2 Hz sharp waves

Your answer is incorrect.

Electroencephalography (EEG) can be useful in differentiating delirium from other conditions. The EEG shows a diffuse slowing of the background rhythm, in patients with delirium. In patients with delirium tremens, the EEG shows low voltage fast activity superimposed on slow waves.

The correct answer is: Low voltage fast activity

Question **25**

Not answered

Marked out of 1.00

Flag question

'Sawtooth wave pattern' in EEG is seen in which of the following phases of sleep?

Select one:

- Stage 1 n-REM sleep
- Stage 3 n-REM sleep
- Stage 2 n-REM sleep
- REM sleep
- Stage 4 n-REM sleep

Your answer is incorrect.

The sawtooth wave is a distinctive pattern in electroencephalogram (EEG) that occurs only in REM sleep and that is recorded mainly in the central scalp area.

The correct answer is: REM sleep

Question **26**

Not answered

Marked out of 1.00

Flag question

Which among the following EEG waves and their frequency is incorrectly matched?

Select one:

- Delta waves- < 4 Hz
- Beta waves- more than 13 Hz
- Lambda- single waves
- Alpha waves- 8 to 13 Hz
- Theta waves- 7-11 Hz

Your answer is incorrect.

The frequency of Theta waves is 4 to 8 Hz. Mu waves occur over the motor cortex and their frequency is between 7-11 Hz

The correct answer is: Theta waves- 7-11 Hz

Question **27**

Not answered

Marked out of 1.00

Flag question

A 33-year-old woman is taking paroxetine for a few months. She developed pins & needles and shock like sensations in her legs. The diagnosis is;

Select one:

- Akathisia
- Serotonin syndrome
- Antidepressant discontinuation syndrome
- Dyskinesia
- Neuroleptic malignant syndrome

Your answer is incorrect.

Antidepressant discontinuation symptoms include the following; Irritability, nausea, ataxia, diaphoresis, paresthesia and increased dreaming. The onset of symptoms is usually within 5 days of stopping treatment and it depends on the half-life of the antidepressant. The risk is increased in those prescribed short half-life drugs such as paroxetine and venlafaxine.

The correct answer is: Antidepressant discontinuation syndrome

Question **28**

Not answered

Marked out of 1.00

Flag question

Which class of drugs should be avoided in organic disinhibition?

Select one:

- Olanzapine
- Benzodiazepines
- Trazadone
- Sodium valpraote
- SSRIs

Your answer is incorrect.

Benzodiazepines should be avoided in organic disinhibition. There is some evidence that SSRIs and Trazadone may ameliorate disinhibition in fronto-temporal dementia

The correct answer is: Benzodiazepines

Question **29**

Not answered

Marked out of 1.00

Flag question

Ms. Triggs has a diagnosis of Bipolar Affective Disorder Type II, with two previous hypomanic episodes and numerous depressive episodes. Her mood is currently low, and she is experiencing insomnia, increased appetite, and negative ruminations. You are considering starting an antidepressant. Which of these is least likely to precipitate a manic switch?

Select one:

- Bupropion
- Venlafaxine

- Sertraline
- Amitryptiline
- Fluoxetine

Your answer is incorrect.

Tricyclics like Amitryptilineand Venlafaxine in particular carry a greater risk of precipitating a switch into mania. SSRIs have also been implicated; there is some evidence for bupropion also contributing to manic switch, but this evidence is less robust than for the other agents listed.

The correct answer is: Bupropion

Question **30**

Not answered

Marked out of 1.00

Flag question

Hyder is a 35-year-old male on your assessment ward. He has a long-standing history of depression for which he takes citalopram. He complains of pain in his leg having slipped and fallen over on the ward the previous morning. He comes to you asking for painkillers. Which of the following increases GI bleeds with SSRIs?

Select one:

- Codeine
- Diclofenac
- Paracetamol
- Ibuprofen gel
- Tramadol

Your answer is incorrect.

SSRIs inhibit the serotonin transporter, which is responsible for the uptake of serotonin into platelets. It is thought that SSRIs deplete platelet serotonin, leading to a reduced ability to form clots and a subsequent increase in the risk of bleeding. Non-steroidal anti-inflammatory drugs, such as ibuprofen and diclofenac, are another class of drug that causes gastrointestinal bleeding (Ref. BMJ 2005).

The correct answer is: Diclofenac

Question **31**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs may exacerbate the symptoms of Schizophrenia due to NMDA receptor antagonistic activity?

Select one:

- Cannabis
- Cocaine
- Ketamine
- LSD
- Amphetamine

Your answer is incorrect.

NMDA receptor antagonists like Ketamine and Phencyclidine have been shown to induce both positive and negative symptoms of Schizophrenia in healthy volunteers. This is possibly due to the modulation of the dopamine system and therefore exacerbates symptoms of patients with Schizophrenia.

The correct answer is: Ketamine

Question **32**

Not answered

Marked out of 1.00

Flag question

Which drug should be used with caution in patients with psoriasis?

Select one:

- Lithium
- Olanzapine
- Haloperidol
- Carbamazepine
- Sodium valproate

Your answer is incorrect.

Lithium is involved in a variety of cutaneous reactions including psoriasis, which may present as exacerbation of preexisting psoriasis, induction of de novo psoriasis, pustular psoriasis, nail changes, and psoriatic arthropathy. The appearance of psoriatic lesions may occur at normal therapeutic serum lithium levels. The refractory period for the development of psoriatic lesions is variable and generally longer in induction and shorter in exacerbation of psoriasis. Lithium-induced psoriasis is often resistant to conventional treatment modalities, and some cases may require dose reduction or discontinuation of lithium treatment. (Ref: Jafferany M. Lithium and Psoriasis: What Primary Care and Family Physicians Should Know. Primary Care Companion to The Journal of Clinical Psychiatry. 2008;10(6):435-439)

The correct answer is: Lithium

Question **33**

Not answered

Marked out of 1.00

Flag question

A 26-year-old woman on psychotropic medications presents to the A&E department in a confused state. On examination, she has fever, raised BP, excessive sweating and bilateral clonus. The most likely diagnosis is;

Select one:

- Hyponatraemia
- Neuroleptic malignant syndrome
- Antidepressant discontinuation syndrome
- Munchausen syndrome
- Serotonin syndrome

Your answer is incorrect.

Serotonin syndrome is a rare but potentially fatal syndrome occurring in the context of initiation or dose increase of a serotonergic agent, characterized by altered mental state (confusion), agitation, tremor, shivering, diarrhea, hyperreflexia, clonus, ataxia and hyperthermia. Autonomic disturbance – namely Hypertension, Tachycardia, Hyperthermia and Excessive sweating are also common.

The correct answer is: Serotonin syndrome

Question **34**

Not answered

Marked out of 1.00

Flag question

Clive is a 27-year-old skateboard artist with paranoid schizophrenia. He was commenced on clozapine two months ago. He attends his GP complaining of severe chest pain. His GP takes recordings of his vitals and performs an ECG. What is likely to be seen on Clive's ECG?

Select one:

- ST depression
- Tall peaked u waves
- Prolonged QT
- Pathological q waves
- ST elevation

Your answer is incorrect.

Clive is most likely suffering from myocarditis, an inflammation of the heart that can be caused by clozapine. ECG findings include: diffuse T wave inversions and saddle-shaped ST segment elevations.

The correct answer is: ST elevation

Question **35**

Not answered

Marked out of 1.00

Flag question

Which of the following is the biggest risk factor for developing NMS?

Select one:

- Afrocarribean ethnicity
- Sudden withdrawal of antiparkinsonian medication
- Older age
- Female gender
- Starting on anticholinergic medications

Your answer is incorrect.

The syndrome of NMS has been described in all age groups. Age is not a risk factor. Men outnumber women 2 fold. Sudden withdrawal of antiparkinsonian medication is one of the high risk factors for developing NMS.

The correct answer is: Sudden withdrawal of antiparkinsonian medication

Question **36**

Not answered

Marked out of 1.00

Flag question

Jack Lyndon is a 43-year-old man with a long history of depression, who one month previously had been started on an antidepressant he's not previously tried. He has found some alleviation in his depressive feelings, his hypersomnia and lethargy. He is invited to a dinner party and part-way through the evening he begins to feel unwell, with facial flushing. The host offers to measure Mr. Lyndon's blood pressure with a portable machine, and it is found to be 208/106. Which of the following is most likely to be related to the hypertensive crisis?

Select one:

- Sertraline
- Fluoxetine
- Mirtazapine
- Agomelatine
- Tranylcypromine

Your answer is incorrect.

When MAOIs such as Tranylcypromine are taken, the patient must be warned to avoid foods (such as mature cheese and red wine) containing tyramine (a monoamine precursor), as it can precipitate a hypertensive crisis, which is potentially life-threatening.

The correct answer is: Tranylcypromine

Question **37**

Not answered

Marked out of 1.00

Flag question

Mrs Davis is a 55-year-old woman who has been taken to A&E by her daughter. She was found to be behaving increasingly oddly over the past few days, isolating herself, suspicious of the neighbours, preoccupied with ideas that she is being persecuted. Further investigation reveals that, following a recent Deep Vein thrombosis, the GP had abruptly stopped one of her usual prescribed medications. Which one of the following prescribed medications is most likely to be implicated in the above scenario?

Select one:

- Androgen
- Testosterone
- Progesterone
- Estrogen
- Thyroxine

Your answer is incorrect.

Hormone Replacement Therapy is frequently given to post-menopausal women. Estrogen is a powerful psychoactive hormone, with widespread physiological effects. It is believed that its psychoactive mechanism is mediated by endorphins. There are reports indicating that sudden withdrawal of estrogens has precipitated mood disturbances and psychosis-like presentations.

The correct answer is: Estrogen

Question **38**

Not answered

Marked out of 1.00

Flag question

George Smith, a 74-year-old retired teacher, presents to your outpatient clinic with his wife, Mary. She has expressed her concern that George has been spending an increasing amount of time in the betting office. His medical history includes; hypertension, Parkinson's disease, depression and gastrooesophageal reflux disease. Of the following medications he is taking, which is the most likely cause of pathological gambling?

Select one:

- Levodopa

- Zopiclone
- Mirtazapine
- Omeprazole
- Metoprolol

Your answer is incorrect.

Impulsive and compulsive behaviour is a possible side effect of some medications used to treat Parkinson's disease, including levodopa. Although rare, it can have a profound effect on the patient and their family. It is important to discuss this risk with patients when prescribing medication for Parkinson's disease. (data from Parkinson's UK)

The correct answer is: Levodopa

Question **39**

Not answered

Marked out of 1.00

Flag question

Sophie had a review in the outpatients' clinic for her depressive disorder. The Psychiatrist is keen to start her on Agolematine. Which of the following baseline investigation should be done before starting her on this drug?

Select one:

- ECG
- Kidney function test
- Liver function test
- Complete blood count
- Thyroid function test

Your answer is incorrect.

Liver toxicity is a side effect of Agomelatine. Liver function test should be performed in every patient before starting him or her on this drug. LFTs should be monitored at 3, 6, 12 and 24 weeks after starting treatment.

The correct answer is: Liver function test

Question **40**

Not answered

Marked out of 1.00

Flag question

Which of the following can be used as an adjuvant to Risperidone to reduce Hyperprolactinaemia?

Select one:

- Sulpiride
- Aripiprazole
- Naloxone
- Trazadone
- Amisulpride

Your answer is incorrect.

Hyperprolactinaemia and related symptoms are reported to improve fairly promptly following the addition of non-prolactin elevating drug such as Aripiprazole.

The correct answer is: Aripiprazole

Question **41**

Not answered

Marked out of 1.00

Flag question

You are discussing clozapine treatment with Luke, a 24-year-old male with treatment refractory paranoid schizophrenia. Which of the following lifestyle factors will produce an increased level of clozapine?

Select one:

- He plans to stop smoking by his next birthday in two months' time
- He drinks on average four pints of lager every week
- He stopped taking fluvoxamine six years ago following a single episode of depression.
- He doesn't drink coffee or tea
- He is taking propanolol for anxiety

Your answer is incorrect.

Clozapine plasma concentrations can rise 1.5 times in the 2–4 weeks following smoking cessation (de Leon, 2004) and in some instances by 50–70% within 2–4 days. The effects of inhibitors (caffeine) are seen sooner than those of inducers (smoking), which require CYP1A2 synthesis (de Leon, 2004).

The correct answer is: He plans to stop smoking by his next birthday in two months' time

Question **42**

Not answered

Marked out of 1.00

Flag question

Which of the following medications acts on NMDA receptors?

Select one:

- Tranylcypromine
- Moclobemide
- Memantine
- Mirtazapine
- Methodone

Your answer is incorrect.

Antidementia medication, Memantine, works by blocking N-methyl-d-aspartate (NMDA) glutamate receptors. Unlike ketamine, which is a high affinity noncompetitive blocker, memantine is a non-competitive blocker with low affinity and binds only to actively open NMDA channels. Its receptor dissociation rate is relatively fast and so it does not accumulate and interfere with normal NMDA activity.

The correct answer is: Memantine

Question **43**

Not answered

Rosy has been diagnosed with generalized anxiety disorder. Which drug is licensed in the treatment of GAD?

Marked out of 1.00

Flag question

Select one:

- Aripiprazole
- Pregabalin
- Lamotrigine
- Topiramate
- Gabapentin

Your answer is incorrect.

Pregabalin is licensed in the treatment of GAD. Several large RCTs have demonstrated its efficacy and tolerability and comparable speed of onset of action to a benzodiazepine.

The correct answer is: Pregabalin

Question **44**

Not answered

Marked out of 1.00

Flag question

You are asked to see a patient in the medical ward. He was clinically depressed and treated on Fluoxetine. His routine investigations showed sodium levels of 122 mol/L. Fluoxetine has now been stopped. Which of the following anti-depressants would be a suitable option in this situation?

Select one:

- Mirtazapine
- Amitriptyline
- Paroxetine
- Moclobemide
- Citalopram

Your answer is incorrect.

This person has developed hyponatraemia (normal range is 136-145 m mol/L) following treatment with antidepressants such as Fluoxetine. For those who develop hyponatraemia with an SSRI, consider Noradrenergic drugs such as Nortriptyline and Lofepramine or an MAOI such as Moclobemide.

The correct answer is: Moclobemide

Question **45**

Not answered

Marked out of 1.00

Flag question

Mrs Swami is about to start taking Olanzapine for a newly diagnosed psychotic illness. Her nephew is reading up on Olanzapine's metabolism and modes of action. What property of a drug such as Olanzapine aids its movement into fat tissues?

Select one:

- It is renally excreted
- It is lipophilic
- It induces Cytochrome CYP450
- It is metabolised in phase I oxidative metabolism
- It is highly protein-bound

Your answer is incorrect.

Lipophilic molecules travel well across the phospholipid bilayer of the cell membrane.They are nonpolarized and nonionized. Greater the lipophilicity, higher the movement of drugs into fat tissues.

The correct answer is: It is lipophilic

Question **46**

Not answered

Marked out of 1.00

Flag question

Which of the following is an aspect of pharmacokinetics ?

Select one:

- Storage release
- Absorption
- Neurotransmitter synthesis
- Neurotransmitter storage
- Reuptake

Your answer is incorrect.

Pharmacokinetics is concerned with the time course and disposition of drugs in the body (what the body does to the drug). It involves the processes of Absorption, distribution, Metabolism and Elimination (ADME). Contrast with pharmacodynamics, which refers to the study of the mechanism of action of drugs (the effect of drugs on the body). Most psychotropics affect neurotransmitters of the brain. This effect can occur at various levels.

The correct answer is: Absorption

Question **47**

Not answered

Marked out of 1.00

Flag question

Shalinee, a 30-year-old florist, has a history of epilepsy for which she takes carbamazepine. She is planning to start a family with her husband, David and seeks your advice on medication during pregnancy. Which of the following pharmacokinetic changes occurs during pregnancy?

Select one:

- Increased plasma protein binding
- Reduced volume of distribution
- Reduced absorption from intramuscular site of injection
- Increased renal elimination
- Reduced rate of metabolism

Your answer is incorrect.

Gastric emptying and small intestine motility are reduced in pregnancy due to elevation of progesterone, which tends to reduce absorption via the oral route. Total body water increases by up to 8 litres, creating a larger space within which hydrophilic drugs may distribute. Total plasma concentration of albumin-bound drugs decreases as a result of haemodilution and the subsequent fall in plasma albumin concentration and reduced plasma protein binding. Some enzymes of the hepatic cytochrome P-450 system are induced by oestrogen/ progesterone, resulting in a higher rate of

metabolism (and hence elimination) of drugs. Renal blood flow is increased by 60-80% during pregnancy, and glomerular filtration rate rises by 50%, leading to enhanced elimination of drugs (Ref. Dawes 2001)

The correct answer is: Increased renal elimination

Question **48**

Not answered

Marked out of 1.00

Flag question

Mr. Welton's memory is being tested. He has been given a list of items to study. Once he has finished studying the items he is given another list to study, before being asked to recall as many items as he can from the first list, in any order. What memory test does this allude to?

Select one:

- Semantic Memory
- Immediate Free Recall
- Working Memory
- Iconic Memory
- Delayed Free Recall

Your answer is incorrect.

Explanation:

Free recall is the retrieval of information from memory without the help of cues, and hence often contrasted with cued recall. This term is also used for retrieval of a number of items of information in any order, in contradistinction to serial recall. In this case, the free recall is also delayed.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Delayed Free Recall

Question **49**

Not answered

Marked out of 5.00

Flag question

Switching antidepressants
What is the advised route for the following changes in medication?

| | |
|---|---|
| Phenelzine to Sertraline | Choose... |
| Trazodone to Mirtazapine | Choose... |
| Fluoxetine to duloxetine | Choose... |
| Amitryptiline to citalopram | Choose... |
| Fluoxetine to Tranylcypromine | Choose... |

Your answer is incorrect.

When swapping from one antidepressant to another, abrupt withdrawal should usually be avoided. Cross tapering is preferred, where the dose of the ineffective or poorly tolerated drug is slowly reduced while the new drug is slowly introduced. 1) Cross tape

The correct answer is: Phenelzine to Sertraline → Withdraw and wait for 2 weeks, Trazodone to

Mirtazapine → Cross taper with caution, Fluoxetine to duloxetine → Stop and start other drug immediately, Amitryptiline to citalopram → Half dose, start new drug, then slow withdrawal, Fluoxetine to Tranylcypromine → Withdraw and wait for 5-6 weeks

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